

✓ 1. DATA EXPLORATION AND PREPROCESSING

```
import pandas as pd
df = pd.read_excel('Airbnb_data.xlsx')
print(df.shape)
df.head()
```

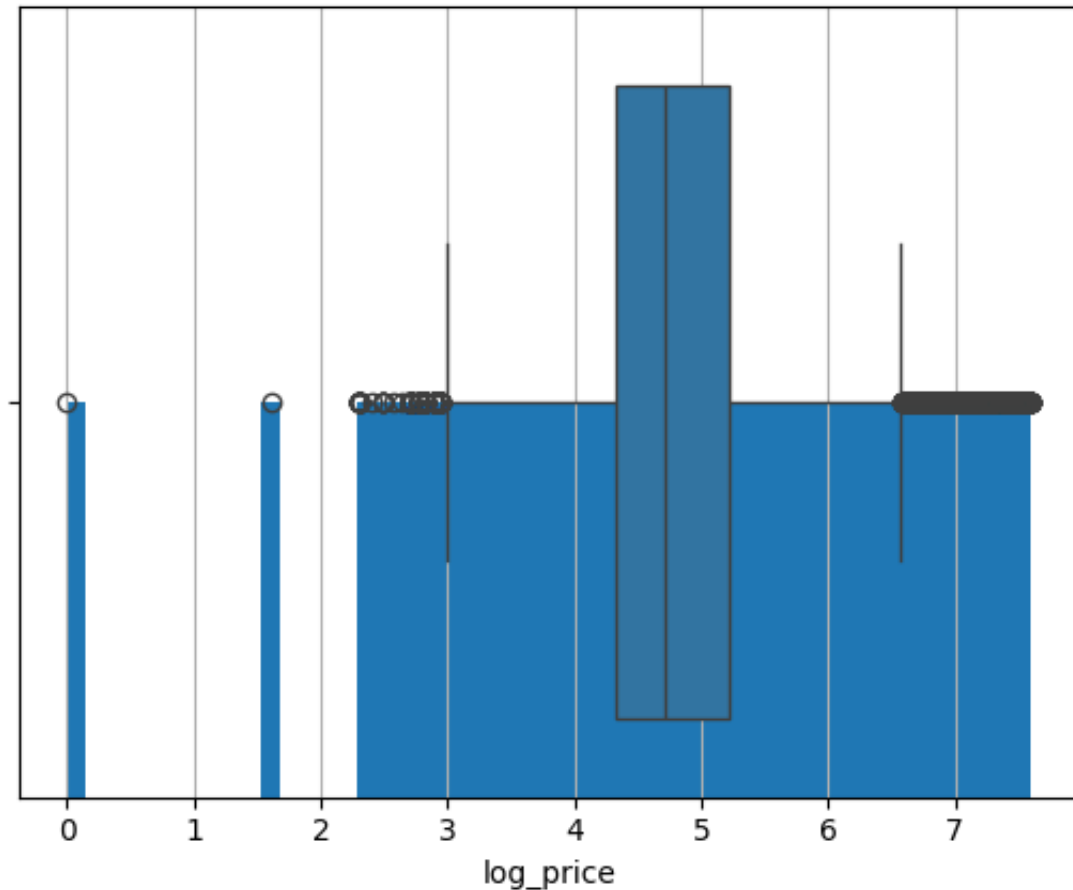
↗ (74111, 29)

	id	log_price	property_type	room_type	amenities	accommoda
0	6901257	5.010635	Apartment	Entire home/apt	{"Wireless Internet","Air conditioning",Kitch...	
1	6304928	5.129899	Apartment	Entire home/apt	{"Wireless Internet","Air conditioning",Kitch...	
2	7919400	4.976734	Apartment	Entire home/apt	{TV,"Cable TV","Wireless Internet","Air condit...	
3	13418779	6.620073	House	Entire home/apt	{TV,"Cable TV",Internet,"Wireless Internet",Ki...	
4	3808709	4.744932	Apartment	Entire home/apt	{TV,Internet,"Wireless Internet","Air conditio...	

5 rows x 29 columns

```
df.describe()
df['log_price'].hist(bins=50)
df.isnull().sum()
import seaborn as sns
import matplotlib.pyplot as plt
sns.boxplot(x=df['log_price'])
```

 <Axes: xlabel='log_price'>



```
df = df.drop(columns=['host_response_rate', 'first_review', 'thumbnail_url', 'l
df = df.dropna()
df = df[['log_price', 'property_type', 'room_type', 'accommodates', 'bathrooms',
        'bed_type', 'cancellation_policy', 'cleaning_fee', 'city', 'bedrooms', 'bec
print("Remaining columns:", df.columns.tolist())
print("Remaining rows:", len(df))
```

 Remaining columns: ['log_price', 'property_type', 'room_type', 'accommodate
Remaining rows: 51545

```
df = pd.get_dummies(df, drop_first=True)
df = df.reset_index(drop=True)

print("DataFrame shape after encoding:", df.shape)
df.head(2) # Show first 2 rows only
```

➡ DataFrame shape after encoding: (51545, 51)

	log_price	accommodates	bathrooms	cleaning_fee	bedrooms	beds	property &
0	5.010635	3	1.0	True	1.0	1.0	
1	5.129899	7	1.0	True	3.0	3.0	

2 rows x 51 columns

```
X = df.drop('log_price', axis=1)
y = df['log_price']

print("Shape of X (features):", X.shape)
print("Shape of y (target):", y.shape)
```

➡ Shape of X (features): (51545, 50)
Shape of y (target): (51545,)

```
X = pd.get_dummies(X, drop_first=True)

print("Shape of X after encoding:", X.shape)

➡ Shape of X after encoding: (51545, 50)
```

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random

print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)
```

➡ X_train shape: (41236, 50)
X_test shape: (10309, 50)
y_train shape: (41236,)
y_test shape: (10309,)

```

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Scaled training data mean (first 5 features):", X_train_scaled.mean(axis=0))
print("Scaled training data std (first 5 features):", X_train_scaled.std(axis=0))

↩ Scaled training data mean (first 5 features): [ 4.81179210e-17  1.45689175e-17
-4.75148316e-17]
Scaled training data std (first 5 features): [1.  1.  1.  1.  1.]

```

✓ 2. MODEL DEVELOPMENT

```

from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train_scaled, y_train)

print("Number of model coefficients:", len(model.coef_))
print("First 5 coefficients:", model.coef_[:5])

↩ Number of model coefficients: 50
First 5 coefficients: [ 0.15834979  0.08681548  0.00329971  0.12553362 -0.0

```

✓ 3. MODEL EVALUATION

```

from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import numpy as np
import matplotlib.pyplot as plt

y_pred = model.predict(X_test_scaled)

mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

# Print Results
print("RMSE:", round(rmse, 3))

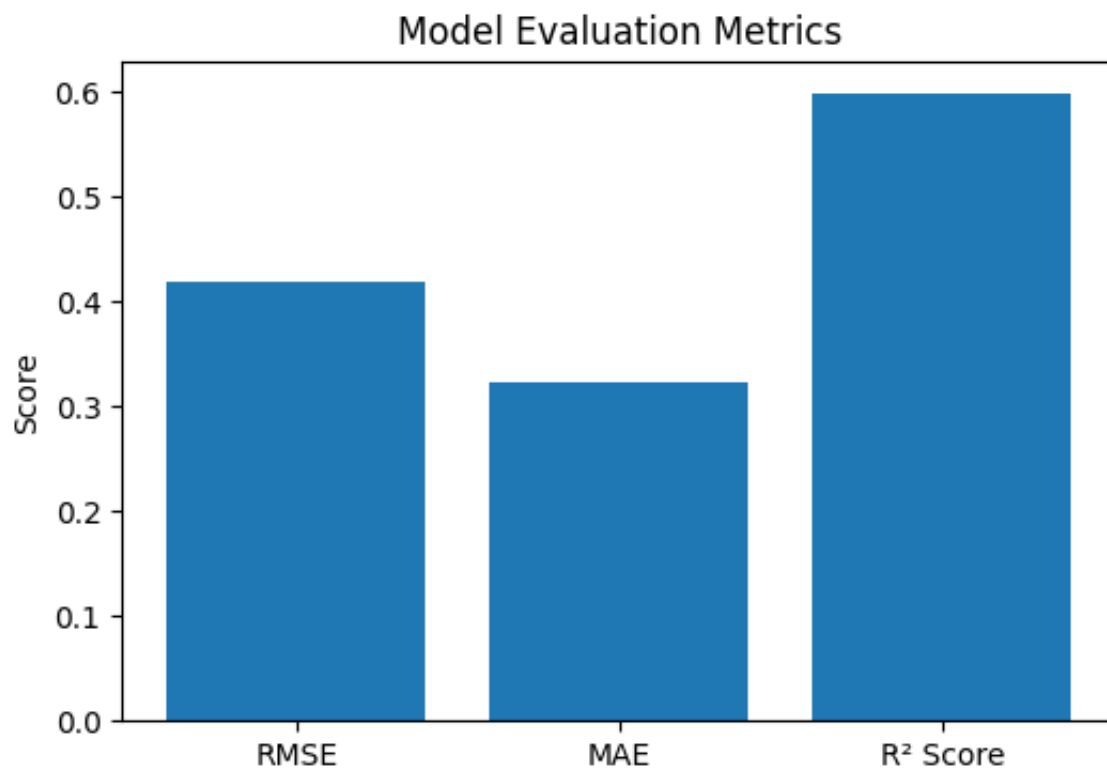
```

```
print("MAE:", round(mae, 3))
print("R2 Score:", round(r2, 3))

# Bar Chart
metrics = ['RMSE', 'MAE', 'R2 Score']
values = [rmse, mae, r2]

plt.figure(figsize=(6, 4))
plt.bar(metrics, values)
plt.title("Model Evaluation Metrics")
plt.ylabel("Score")
plt.show()
```

RMSE: 0.418
MAE: 0.322
R² Score: 0.598



Google Drive Link to project Video Explanation, Presentation , and Code Files:

https://drive.google.com/drive/folders/1ev3ABZfVt1-Noy_bMUXaUK-82nHUpIO3?usp=sharing

